Phase 3 | Project Design Phase

Proposed Solution

AI-Powered Travel Planner | Smart Situational Itinerary Engine

Live URL:	https://travel-planner-iota-plum.vercel.
GitHub:	github.com/itskian2007/travel-planner
Tech:	React + FastAPI + PostgreSQL + Vercel
Team:	itskian2007

1. Solution Summary

The AI-Powered Travel Planner is a full-stack web application that generates personalised 10-day travel itineraries based on user preferences. The solution combines expert-curated Day 1-3 content with a procedural engine for Days 4-10, covering 20 global destinations with 304 verified landmark images. Live at:
https://travel-planner-iota-plum.vercel.app/

2. Core Solution Features

20 Destinations

Tokyo, Bali, Rome, Paris, Iceland, Swiss Alps, Kyoto, Barcelona, Bangkok, Dubai, Seoul, New York, London, Amsterdam, Singapore, Sydney, Cairo, Istanbul, Rio, Cape Town

5 Filter Dimensions

Mood: Adventure/Cultural/Relaxation/Romance
Budget: Budget/Mid-Range/Luxury
Diet: Veg/Vegan/Halal/Non-Veg/Gluten-Free
Persona: Solo/Family/Couple/Group
Accessibility Preferences

10-Day Itinerary

Day 1-3: Expert-curated landmark sequences
Day 4-10: Procedural engine generates unique activities
Each day: 3-4 attractions + dining + local tips
All with verified Wikipedia landmark images

User Authentication

Secure registration with email/password
JWT-based session management (24hr tokens)
Forgot password with email reset flow
Personalised preference persistence

304 Unique Images

Zero duplicate images across all destinations
All sourced from Wikipedia verified uploads
High-resolution landmark photography
Progressive loading for performance

Live Deployment

Hosted on Vercel (global CDN edge network)
FastAPI backend on cloud infrastructure
PostgreSQL managed database
99.9% uptime SLA

3. Solution Differentiation

Capability	Our Solution	Typical Travel Apps	Advantage
Personalisation Depth	**5 dimensions**	1-2 filters	**Deep personalisation**
Itinerary Days	**10 days auto-gen**	3-5 days manual	**2x more coverage**
Destination Count	**20 global cities**	5-10 cities	**Broader choice**
Image Authenticity	**304 Wikipedia-verified**	Stock images	**Zero fake images**
Diet Support	**5 diet types**	None/generic	**Inclusive design**
Accessibility	**Filter supported**	Not available	**WCAG compliant**
Cost	**Free (open source)**	Freemium/Paid	**Zero barrier**
Deployment	**Vercel global CDN**	Single server	**Global availability**